

ORIGINAL ARTICLE

Development and evaluation of artificial intelligence literacy training for teacher education students

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Abstract

Teacher education students play double role as present learners and future educators. Hence, they need targeted training to navigate the growing influence of Generative Artificial Intelligence (GenAI) on teaching, learning and professional identity. However, existing artificial intelligence (AI) literacy programmes predominantly emphasize technical AI knowledge and pre-GenAI tools or are offered by GenAI platforms that focus on their own technologies' features, thereby lacking pedagogically structured frameworks that address non-technical dimensions such as ethics, contextual understanding and the development of human-centered, critical and lifelong learning mindsets to adapt to rapid technology changes. Furthermore, limitation of time in teacher education syllabus and faculty's lack of AI literacy require targeted intervention. This design-based research (DBR) aims to fill this gap by developing and evaluating a set of design principles for GenAI literacy training in teacher education. Integrating contemporary AI competency frameworks for learners and teachers, the study implements these principles in a workshop that serves as the research prototype. The prototype was first piloted with 14 master's students and then evaluated with 29 teacher education (TE) students. Results show significant gains in participants' AI competence self-efficacy, attitudes towards GenAI and commitment to critical, ethical and pedagogical

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engagement with GenAI tools. The findings highlight the need for teacher education programmes to integrate GenAI literacy that supports teachers' evolving roles as reflective practitioners, co-creators and life-long learners in an AI-driven world.

KEYWORDS

AI competency, AI literacy, Design-based research, Human–AI collaboration, Human-AI shared regulation of learning, Hybrid Intelligence, Teacher education students

Practitioner notes

What is already known about this topic:

- Students' self-efficacy and mindset influence their adoption of GenAI in learning and teaching. GenAI competency is essential for teacher education students, encompassing technological, ethical and sociocultural dimensions.
- Implementing AI literacy training faces challenges such as educators' limited AI literacy, the lack of pedagogical frameworks and the need to integrate technical knowledge with ethical reasoning, self-regulated learning and metacognitive skills. Transformative learning is crucial in preparing teacher education students as co-creators and facilitators in AI-enhanced education.

What this paper adds:

- This study presents the design and evaluation of an AI competency training programme to enhance teacher education students' self-efficacy, critical and human-centered AI mindsets and human–AI interaction skills.
- Using design-based research, it identifies key principles for AI literacy training, demonstrating how transformative, active and experiential learning approaches can prepare teacher education students for evolving roles in AI-enhanced education.

Implications for practice and/or policy:

- Educational technologists can apply the proposed design principles to develop GenAI tools and AI systems that enhance AI competency and self-regulated learning (SRL).
- Educators can integrate these AI training principles into their teaching strategies and enhance their task assignments and assessments to foster deep learning, self-regulated learning and metacognitive skills in the age of GenAI.
- Policymakers should update teacher education curriculum to include AI competency, strengthen content and pedagogical knowledge, especially in social emotional learning, formative and authentic assessment methods and establish guidelines for selecting GenAI tools that support human intelligence and mitigate over-reliance.

INTRODUCTION

Artificial intelligence (AI) has become increasingly common in education following the 2022 release of generative AI (GenAI) tools such as ChatGPT and Midjourney. These technologies are expected to reshape future employment landscapes, with automation projected to create millions of new roles while transforming many existing ones within 5 years (World Economic Forum [WEF], 2025). This technology puts pressure on preparing learners today for resilience and adaptability as approximately 65% of today's primary students will enter currently non-existent jobs (Eswaran, 2024). Unlike earlier technological disruptions, GenAI introduces complex ethical, pedagogical and cognitive challenges that demand new forms of competence for both learners and educators (Cukurova, 2025; Molenaar, 2022b; O'Dea & Ng, 2024). The European Union's AI Act (EU's Regulation 2024/1689, n.d.) further underscores this urgency by mandating adequate AI literacy for those engaging with high-risk AI systems, including those used in educational settings. Consequently, teacher education (TE) students require targeted preparation to navigate GenAI both as learners—during their transformative higher education years—and as future educators of a 'GenAI-native' generation.

Despite policy and research attention, AI literacy training remains inadequate in teacher training programmes, intensifying the risks of misuse and limiting the potential of AI application for innovative pedagogy or learning enhancement (Gurl et al., 2024; Sperling et al., 2024). Teacher training programmes, like other non-technical disciplines, also face curriculum constraints and limited AI literacy among faculty (O'Dea & Ng, 2024; Sperling et al., 2023). While studies suggest that TE students' familiarity with AI tools, attitudes and mindsets directly influence their willingness to explore and integrate AI into their teaching and learning practices (Hsu et al., 2024; Nguyen, Shalpegin, et al., 2024), existing AI literacy curricula in K12 and higher education predominantly focus on the technical aspects and application of rule-based AI technology and have not been updated to include GenAI tools (Almatrafi et al., 2024; Chee et al., 2024; Ng, Leung et al., 2021; Sperling et al., 2024).

Beside curriculum imbalance, there is a lack of an effective pedagogical approach for non-technical aspects such as ethics or mindset in AI application. Sperling et al.'s (2024) review revealed that ethical considerations in AI literacy programmes are often limited to understanding algorithmic decision-making or AI system configurations, neglecting broader human agency, educational, societal, economical and environmental implications of current GenAI technology. Existing initiatives to update AI literacy with GenAI application, mainly done by for-profit organizations (such as Google's GenAI for Educators), also emphasize instruction of prompt engineering on their systems. This approach limits the development of critical thinking and evaluation skills of diverse GenAI solutions (United Nations Educational, Scientific and Cultural Organization [UNESCO], 2023). Additionally, self-paced online learning formats restrict opportunities for active learning and primarily engage students who are already highly motivated or have surpassed affective barriers (Terras & Ramsay, 2015). Without fostering critical reflection, metacognitive skills and human–AI interaction skills, AI literacy training may have limited impact on TE students' long-term professional development and lifelong learning engagement, risking obsolescence with rapid AI advancements.

In response to these challenges, this study adopts a design-based research (DBR) approach to explore: *How can AI literacy training be effectively implemented for TE students?* The study aims to formulate a set of design principles that enhance TE students' self-efficacy and AI competency, preparing them both as learners and future educators. Through iterative cycles, the study conceptualizes design principles grounded in literature and frameworks such as UNESCO's (2024a, 2024b) AI competency frameworks for students and teachers, TPACK (Mishra et al., 2023) and the Human–AI Shared Regulation for Learning (HASRL) model (Järvelä et al., 2023). An introductory AI literacy workshop was developed as the

research artefact to operationalize and evaluate these principles. Its impact was assessed through TE students' AI competence self-efficacy, reflective practices and ability to apply GenAI in lesson planning and learning material creation. This study contributes to on-going discourse on human–AI interaction in learning from students' perspectives and offers practical implications for AI literacy training in TE programmes.

THEORETICAL BACKGROUND

GenAI in teaching and learning: Opportunities and challenges

Generative AI (GenAI) is a subset of artificial intelligence (AI) technology that can produce new content in various formats with intuitive interface and human-like interaction from users, making it easily accessible by non-STEM users (Bozkurt, 2024). Increasing access to GenAI accelerates the adoption of AI in education, offering both transformative opportunities and unprecedented challenges (Järvelä et al., 2024; Nguyen et al., 2023; O'Dea & Ng, 2024).

GenAI can enhance teaching efficiency by supporting lesson planning, content delivery, assessment design, research and automating administrative tasks (Chiu, 2024; Chiu, Ahmad, Ismailov, et al., 2024; Rejeb et al., 2024; Tlili et al., 2023). It fosters adaptive learning environments for students, such as AI-powered chatbots facilitating debates (Guo et al., 2023) or serving as virtual mentors to simulate teaching practices for teacher education (TE) students (Son et al., 2024). However, these advancements necessitate a reconfiguration of teaching roles, requiring educators to navigate GenAI's pedagogical, ethical and classroom implications and learners to be active and adaptive (Zhai, 2024).

Despite its potential, GenAI integration carries critical risks when AI literacy is underdeveloped in TE programmes. Key concerns include overdependence, ethical issues related to academic integrity, data privacy and intellectual property (Nguyen et al., 2023; O'Dea & Ng, 2024). Research highlights risks such as 'metacognitive laziness', diminished intrinsic motivation (Fan et al., 2025) and reduced learning outcomes when students overly rely on GenAI tools (Bastani et al., 2024; Darvishi et al., 2024; Hill, 2023). Moreover, faculty often misjudge AI's capabilities, leading to delayed pedagogical adaptation, reliance on unreliable AI-detection tools and attempts to 'AI proof' assessments instead of leveraging GenAI to support learning (O'Dea & Ng, 2024). Without structured AI literacy training, TE students report anxiety and uncertainty, struggling to critically evaluate or meaningfully integrate GenAI into learning and teaching (Bae et al., 2024; Hsu et al., 2024). Effective AI literacy programmes are essential to prepare TE students as adaptive, reflective learners and ethical facilitators in AI-enhanced classrooms.

Relevant frameworks for AI literacy training of TE students

AI literacy frameworks

Despite growing interest in AI in education, AI literacy frameworks tailored specifically to TE students remain limited (Almatrafi et al., 2024; Chee et al., 2024; Sperling et al., 2024). To inform the design of AI literacy training for TE students, this study reviewed existing frameworks targeting non-technical students and educators. Long and Magerko (2020) define AI literacy as competencies that enable individuals to critically evaluate, communicate and apply AI across contexts. Ng, Leung et al. (2021) structure AI literacy in four dimensions: know and understand, use and apply, evaluate and create AI. O'Dea et al. (2024) built upon this model to GenAI, emphasizing competencies for ethical reasoning in use, apply and evaluate GenAI. Chiu,

Ahmad, Ismailov, et al. (2024) define AI literacy as foundational knowledge and skills, while AI competency expands to reflective, ethical and confident application. Similarly, Ng, Wu et al. (2024) add affective, behavioural, cognitive and ethical dimensions, underscoring self-efficacy and higher-order thinking for responsible AI use. Chee et al. (2024) argue that AI literacy for HE learners should move beyond basic conceptual knowledge and rule-based ethics to embrace complex ethical dilemmas and creative problem-solving for workplace readiness. However, these frameworks lack instructional context application for TE students as future educators.

UNESCO's AI competency framework and TPACK

UNESCO's (2024a, 2024b) AI competency framework for students and teachers advances a human-centred, critical approach to AI in education. It outlines five interrelated dimensions: human-centered mindset (AIHu), Ethics of AI (AIE), AI understanding and application (AIK) for both students and teachers, AI pedagogy (AIP) and AI for professional development (AIPD) specifically for teachers. The dimensions are structured across three levels of progression: understand, deepen and create. This aligns with calls for future teacher roles and agency transformation and lifelong learning in AI-enhanced educational contexts (Zhai, 2024). However, it offers limited guidance on pedagogical strategies or integration models.

In this regard, the Technological Pedagogical Content Knowledge (TPACK) framework (Mishra & Koehler, 2006) provides useful insights. Effective AI literacy training should integrate technological (TK), pedagogical (PK) and content knowledge (CK). Recent scholarship has extended the TPACK framework to encompass AI technology and the broader ethical, social and cultural dimensions of digitality. Emerging models such as Intelligent-TPACK (Celik, 2023), AI-TPACK (Ning et al., 2024) and the expanded Contextual Knowledge (XK) construct proposed by Mishra et al. (2023) highlight the importance of ethical and socio-cultural sensitivity in the integration of GenAI within classroom practice. Researched TPACK development pedagogy approaches include technology hands-on workshops, microteaching, project-based learning and TPACK-informed lesson design (Lachner et al., 2021; Zhang & Zhou, 2023). However, pedagogical approaches for contextual and ethical AI competency, such as critical evaluation of AI outputs when interacting with GenAI tools, are still not addressed in TPACK pedagogy researches.

Human–AI shared regulation in teaching and learning

Before becoming educators capable of integrating GenAI into their teaching, TE students undergo a transformative process of becoming reflective, lifelong learners while simultaneously acquiring subject content, pedagogical and sociocultural knowledge. Cultivating self-regulated learning in both individual and collaborative contexts is therefore essential (Järvelä et al., 2016; Malmberg et al., 2015; Panadero, 2017; Zimmerman, 2002). Although GenAI can enhance self-regulated learning via adaptive feedback, it can also cause cognitive overload or over-reliance depending on learners' self-regulatory strategies, motivation and AI system design (Hill, 2023; Molenaar, 2022a). Hence, we adopt the HASRL model by Järvelä et al. (2023), which conceptualizes learning regulation as an iterative process in which humans and AI systems jointly enact and adapt cognitive, motivational and socio-emotional regulatory processes.

The center of HASRL is that an adaptive learner model is essential for effective human–AI shared regulation. The framework emphasizes continuous modelling of learners' cognitive, motivational and socio-emotional states to enable AI systems to provide timely and

appropriate support. Importantly, this learner model must track dynamic changes in both learners' regulatory processes and the evolving behaviour of the AI model itself, supporting alignment between human agency and AI adaptivity (Järvelä et al., 2023). In other words, this model highlights the importance of learners' agency in planning, monitoring and controlling their AI-mediated learning. Through ongoing reflection on their regulatory decisions and learning outcomes, students develop stronger self-regulated learning skills. At the same time, AI systems iteratively respond to learners' inputs through detect-diagnose-act-learn cycles, refining the relevance of their feedback.

In practice, early empirical research has suggested that learners who actively engage in goal setting, refining prompts and verifying GenAI outputs, particularly in higher-order tasks like academic writing and reflective lesson planning, have better performance (Gurl et al., 2024; Nguyen, Hong et al., 2024; Nguyen, 2025; Nyaaba et al., 2024). As the GenAI technology, pedagogical knowledge, contextual knowledge (XK) or assessment know-how are still under exploration, HASRL conceptualizes how TE students can simultaneously develop self-regulated learning, professional expertise and AI competencies while interacting with AI systems. Reflection and active learner agency in HASRL underpin the integrated conceptual model of AI literacy training shown in Figure 1.

An integrated approach to AI literacy training

Given the rapid and fragmented evolution of AI literacy research, there is a need to synthesize existing frameworks to provide a coherent foundation for designing effective AI literacy training. Foundational models conceptualize AI literacy as involving conceptual

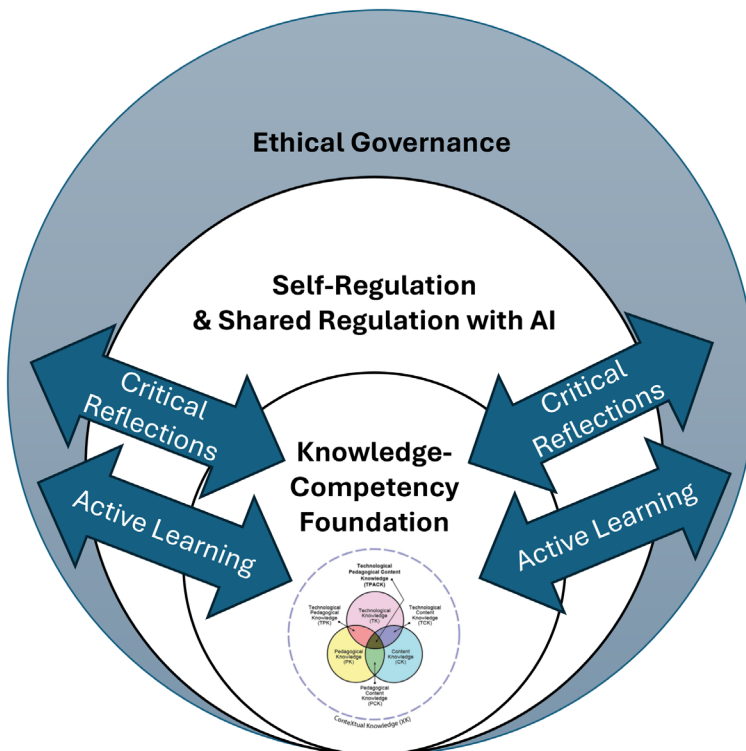


FIGURE 1 Integrated conceptual framework for AI literacy training for teacher education students.

understanding, practical skills and ethical awareness (Long & Magerko, 2020; Ng, Chu, et al., 2021), but they focus primarily on individual cognition and offer limited guidance for pedagogical application. More recent frameworks (Chee et al., 2024; Chiu, Ahmad, Ismailov, et al., 2024; O'Dea et al., 2024; Ng, Tan et al., 2024) extend AI literacy towards GenAI use, data literacy and collaborative competencies, yet these contributions remain loosely connected to instructional design and teacher preparation. UNESCO's AI Competency Framework adds a humanistic and ethical dimension, emphasizing fairness, inclusion and sustainability, but lacks detailed pedagogical strategies for classroom implementation. TPACK provides pedagogical and contextual coherence by showing how teachers can integrate AI meaningfully within subject domains, yet it does not fully capture TE students' regulatory engagement with AI systems as developing learners. Integrating the HASRL model (Järvelä et al., 2023) addresses this gap by foregrounding socio-cognitive processes of co-regulation, shared monitoring and metacognitive adjustment between humans and AI. While TPACK and UNESCO's AI competency framework specify what TE students should know and be able to do with AI, HASRL clarifies how they must learn with, regulate and adapt alongside AI systems whose affordances continue to evolve and whose long-term implications remain uncertain. Taken together, these frameworks offer an integrated perspective that links conceptual, pedagogical, ethical and regulatory dimensions of AI literacy. As illustrated in Figure 1, the synthesis provides the theoretical foundation for design principles guiding AI literacy training. At the core, the model extends TPACK to include GenAI-specific capacities, limitations and appropriate use, requiring teachers to curate and evaluate AI tools responsibly rather than apply them generically—aligning with UNESCO's AI pedagogy and understanding dimensions and emerging GenAI literacy models (Chiu, Ahmad, Ismailov, et al., 2024; Ng, Wu et al., 2024). The second ring foregrounds self- and shared regulation from HASRL, emphasizing continuous planning, monitoring and reflection for adaptive professional development. Finally, ethical governance anchors the model through UNESCO's human-centered and ethical principles, including critical scrutiny of outputs, bias awareness, privacy responsibility and societal and sustainability considerations. Through iterative cycles of participation and critical reflection across knowledge, pedagogical use, regulation and ethics, TE students can develop AI competencies holistically over time.

DESIGN-BASED RESEARCH METHODOLOGY

Design-Based Research (DBR) rooted in design science research, is 'a systematic but flexible methodology aimed to improve educational practices through iterative analysis, design, development, and implementation, based on collaboration among researchers and practitioners in real-world settings' (Wang & Hannafin, 2005, p. 6). DBR is an effective approach to address complex, open educational problems, such as emerging or evolving subject knowledge, undefined pedagogical content knowledge, lack of instructional materials, insufficient teachers' knowledge and complex policy or societal factors (Barab & Squire, 2004; Hoadley & Campos, 2022). Given the rapid evolution of GenAI, and the absence of structured AI literacy pedagogical methods, and lack of faculty AI literacy, DBR provides a suitable, iterative and context-sensitive approach for this study.

Research process

This study employed Reeves's (2006) Design-Based Research (DBR) framework, adapted to the context of AI literacy in teacher education. The process comprised four iterative phases (illustrated in Figure 2): (1) analysing challenges in integrating AI literacy with focus

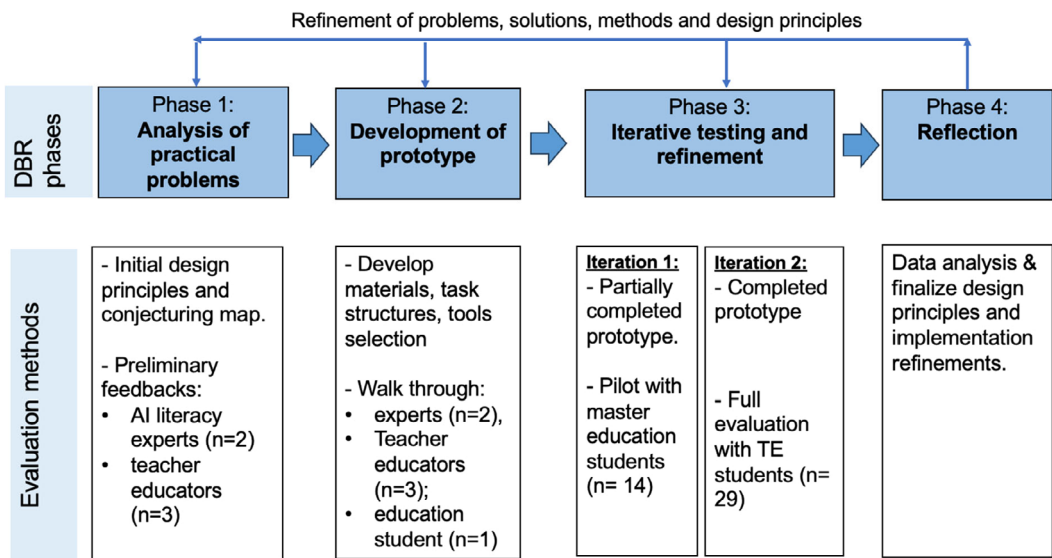


FIGURE 2 Process model of DBR adapted from Reeves (2006) and formative evaluation methods from Plomp (2013).

on GenAI through literature review and teacher educators consultation; (2) developing preliminary design solutions informed by ICT pedagogy and AI literacy frameworks; (3) refining these solutions through two workshop iterations—an initial pilot with 14 master's students and a second prototype with 29 teacher education (TE) students; and (4) synthesizing findings to produce final design principles for effective implementation within TE programmes.

Each phase was continuously formatively evaluated using mixed methods and guided by quality criteria for educational design, including validity, practicality and effectiveness (Plomp, 2013). Based on these evaluations, the research team engaged in iterative reflection and refinement of both the design principles and the conjecture map—a document that links theoretical constructs and design principles to their embodiment in prototypes and anticipated learning outcomes (Sandoval, 2013). This process ensured the research rigour of the DBR methodology (Cotton et al., 2009; Hoadley & Campos, 2022; Sandoval, 2013). (See more in Figure 2).

Participants and contexts

For demonstration, testing and evaluation purposes, two iterations of DBR were conducted with separate cohorts of participants. The first iteration for piloting was conducted with 14 graduate education students (11 female and 3 male) to refine its content, structure and task design. Participants reported moderate use of GenAI tools ($M=3.48$, $SD=0.92$) and moderate engagement in AI training ($M=2.48$, $SD=1.17$) on a 1–5 scale, normalized from the original 1–7 scale using a min–max scaling equation (De Amorim et al., 2022, p. 6).

Second iteration for evaluating the design principles involved 29 final-year TE students (26 female, 3 male, aged 20–22), all in final year with secondary and high school classroom teaching practice experience. The AI literacy training was a pilot required session within the teacher education extracurricular studies programme. Reflecting global trends in the teaching profession, most participants were female (OECD, 2017; TTF, 2017). Participants reported a moderate frequency of GenAI use and experience ($M=2.75$, $SD=1.03$), with most indicating minimal prior exposure to GenAI training or updates ($M=1.69$, $SD=0.98$) on

the same 1–5 scale normalized from the 1–7 scale (De Amorim et al., 2022, p. 6). In total, 27 participants completed both pre- and post-surveys, and 24 submitted final artefacts.

Data collection and measurement

Participants in both iterations were sent an online pre–post workshop survey that included background questions and two Likert scales to measure TEs' GenAI usage frequency as well as their experience and frequency of updating knowledge about AI technology or participating in AI-related training (1 = never to 7 = very frequently).

The effectiveness of the prototypes of AI literacy training was assessed using multiple instruments (see [Appendix 2](#)): (1) the validated Teacher AI Competence Self-Efficacy Scale (TAICS; $\alpha = 0.971$; Chiu, Ahmad, & Çoban, 2024), (2) an adapted training evaluation measure (Markus et al., 2024) examining engagement, difficulty and mental effort, reflective surveys based on Rolfe et al. (2001) and (3) TE students' AI-enhanced lesson plans rubric based on TPACK (Karakaya, 2017; Koh, 2013) (See [Appendix 3](#)). To understand TEs' cognitive and affective impact from the training prototype, we also collect open-ended post-workshop reflections questions and affective responses right after each module. The reflection and affective responses debriefing prompts could be found in [Appendix 1B](#).

Data analysis

For the first iteration, as the prototype was only partially completed and the sample size was small ($n = 14$), only post-workshop reflections and affective responses were analysed. Answers to reflection prompts were analysed using thematic analysis (Bree & Gallagher, 2016), and short affective responses were examined using content analysis (Krippendorff, 2018; Mayring, 2014). The emotional reactions were coded by two researchers with substantial Cohen's Kappa interrater reliability ($\kappa = 0.69$), categorized as positive (e.g. 'excited', 'inspired') or negative (e.g. 'anxious', 'confused') to evaluate training practicality and inform subsequent refinements.

For the second iteration, the effectiveness of the full AI literacy training prototype was evaluated using mixed methods to analyse multiple instruments. A Wilcoxon signed-rank test with Benjamin–Hochberg correction was applied to assess changes in the AI Competence Self-Efficacy Scale (TAICS) (Chiu, Ahmad, & Çoban, 2024), controlling for potential false discovery associated with multiple subscale comparisons. Descriptive statistical analysis was used for the remaining instruments, including the training evaluation measure (Markus et al., 2024) and the TE students' AI-enhanced lesson plan rubric (see [Appendix 3](#) for details). Data from post-workshop open-ended reflections were analysed using thematic analysis and affective responses were coded by the same two researchers and analysed using the same content analysis procedures as applied in iteration 1.

INITIAL DESIGN PRINCIPLES AND DESIGN PROTOTYPE

Initial design principles

The core outcomes of this AI literacy training for TE students focus on affective, behavioural learning and ethics (Ng, Wu, et al., 2024), as well as self-efficacy in AI competencies relevant to the TE students context (Chiu, Ahmad, & Çoban, 2024). These foundations aim to equip TE students with the confidence to engage in ongoing learning, experimentation

and co-creation of AI's role in education. By emphasizing transferable competencies over technical knowhow, the training supports sustained impact amid fast-evolving technologies. Given the emerging nature of AI literacy pedagogy and methods, we developed a set of design principles informed by literature on digital competence, TPACK training (Wang et al., 2014), AI literacy frameworks (Chee et al., 2024; Long & Magerko, 2020; Ng, et al., 2021) and HASRL model (Järvelä et al., 2023). These initial principles were preliminarily checked for early feedback on relevance and consistency (Plomp, 2013) by two AI literacy experts and three teacher educators ($n=3$). The initial principles are conceptualized as follows:

Principle 1. Critical reflections opportunities—training needs to provide opportunities for TE students to critical reflect their roles, agency and impact of GenAI on their learning and education.

Monitoring, self-regulatory process and adaptation is an important part of self-regulated learning and human–AI shared regulation for learning (Järvelä et al., 2023). Beside reflection and adapt on learning outcomes and strategies, to prepare TE students for ethical and pedagogical use of AI, training must incorporate structured opportunities for critical reflection. TE students need space to examine their assumptions about their roles, educational philosophies and the broader implications of GenAI for teaching and learning. Reflection is a core component of self-regulated learning, enabling learners to assess their performance, make attributions and build self-efficacy (Raković et al., 2022). This aligns with UNESCO's (2024b) emphasis on cultivating ethical awareness through deeper reflection on AI's societal impact. Drawing on Mezirow's (2000) transformative learning process, reflection begins with a 'disorienting dilemma', prompting individuals to question existing beliefs (Grocott, 2022; Schols, 2012). Empirical studies highlight that integrating technology with reflective practice can shift TE students from teacher-centered roles towards becoming facilitators of learning (Benitt et al., 2019; King, 2002). Guided by this principle, our training design includes participatory tasks, speculative technology demonstrations and iterative reflection cycles. These are intended to help TE students critically assess their role in AI-enhanced classrooms and cultivate habits of lifelong learning (Zhai, 2024).

Principle 2. Active learning approach—training needs to integrate active learning to build self-efficacy and technological pedagogical knowledge.

Active learning approaches are essential for building TE students' confidence and understanding of AI integration. Reviews of AI literacy programmes emphasize hands-on, collaborative tasks, such as workshops, simulations and AI-based artefacts, as effective for making AI concepts accessible and meaningful (Almatrafi et al., 2024; Long & Magerko, 2020). Active learning supports self-regulated learning by placing learners in control of their goals and engagement (Pinski et al., 2023; Pinski & Benlian, 2024). This method has been successfully applied in some AI literacy programmes in higher education for non-technical students (Laru et al., 2024; Shalpegin & Nguyen, 2024). Moreover, integrating such strategies supports the development of TPACK, aligning AI tools with pedagogical purposes while addressing ethical concerns (Celik, 2023).

Principle 3. Design for Human–AI shared regulation processes—Training needs to focus on developing students' self-regulation and prompting skills to enable reflective, responsible and effective interaction with AI systems.

Effective human–AI interactions in learning require both self-regulation and AI system communication skills. As GenAI tools become more prevalent in classrooms, TE students must learn how to interact meaningfully with these systems. Prompt engineering—the ability to craft, evaluate and revise prompts or requests and input to GenAI—has emerged as a critical skill for both learners and educators (Knoth et al., 2024; Molenaar, 2022b). To support this, our training integrates the prompt engineering concepts for educators (Park & Choo, 2024), guiding TE students to identify needs, design prompts, evaluate outputs and apply them responsibly. Ethical and contextual considerations are embedded, helping TE students discern when and how to use AI effectively. Research shows that targeted interventions in prompt engineering enhance students' self-regulated learning and AI self-efficacy (Lo, 2023; Mzwri & Turcsányi-Szabo, 2025), reinforcing the importance of this principle in fostering adaptive, reflective and responsible AI use.

AI literacy training workshop for TE students as an instantiated research prototype

The instantiated research prototype embodied the design principles by translating theoretical constructs into tangible features integrated within the AI literacy training workshop. We developed a conjecture map (See Figure 3), drawing on Hoadley and Campos (2022) and Sandoval (2013) conjecturing map framework, to ensure logical alignment between design principles, implementation map elements and intended learning outcomes. To align with three design principles, we curated GenAI tools (e.g. text, image, video, slides, infographics) selected for accessibility, privacy, security and suitable cost for TE students. Because GenAI evolves rapidly, the training emphasizes developing a mindset and self-regulatory skills for selecting and engaging with AI,

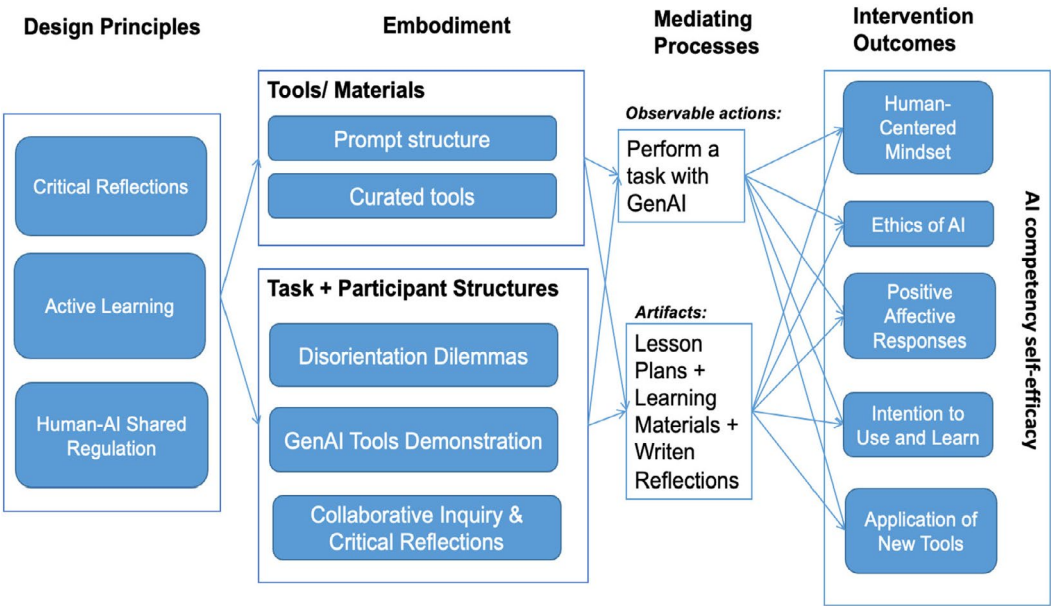


FIGURE 3 Initial conjecturing map applied from Sandoval (2013).

rather than prescribing specific tools. A full tool list and task structures are provided in [Appendix 1A](#) and [1C](#).

The prompt structure (adapted from Lo, [2023](#)) and learning activities incorporated case-based disorientation dilemmas, technology demonstrations and ethical scenarios. Students applied prompt-engineering through guided hands-on tasks and individual/group reflections, aiming to build AI self-efficacy, positive engagement and intention to apply GenAI in teaching and learning.

Informed by the conjecturing map above, the AI literacy training intervention's training materials and detailed task structures were co-developed with feedback from two AI literacy experts, three teacher educators and one education student. The prototype training is delivered in two modules. Each module lasted 2 h, with an additional hour in the final iteration allocated for hands-on assignment work. This modular format ensured feasibility for integration into teacher training programmes and delivery by non-technical teacher educators.

Module 1, '*Teachers' New Mindsets for the AI Era*', addressed AI human-centered mindset, AI pedagogy, ethics and professional development (UNESCO, [2024b](#)). It also aligned with the TPACK framework, contextual knowledge (XK) (Mishra et al., [2023](#)). The module implemented Design Principle 1 through reflective learning cycles: (1) AI demonstrations and case dilemmas; (2) individual reflection; (3) group dialogue using guided prompts; and (4) end-of-session reflection (Grocott, [2022](#); Mezirow, [2000](#)).

Module 2, '*Collaborating with GenAI Tools in Teaching*', focused on the demonstration of GenAI applications in learning, assessment, lesson planning and content creation. Aligned with AI knowledge and application, AI pedagogy, Human-centered mindset dimension and Design Principles 2 and 3, the session included computational thinking strategies, task delegation discussions, guided tool demonstrations, prompt experimentation, AI output checklists and reflections. Participants concluded by submitting a lesson plan and multimedia teaching resources created with GenAI tools. Examples of tasks in each module and alignment with learning outcomes could be found in [Appendix 1C](#) and [1D](#).

DEMONSTRATION AND EVALUATION OF THE AI LITERACY TRAINING PROGRAMME

Iteration 1: Pilot study

Content analysis of short reflections after two modules revealed that 87.1% of participants expressed positive emotional responses to the workshop, including feelings of curiosity, inspiration and excitement. In contrast, 12.9% reported concerns or anxiety, particularly about AI's long-term societal impact (See [Appendix 4A](#)). When participants identified their key takeaways from the workshop, the inductive coding scheme from King ([2002](#)) revealed shifts in their perspectives. Half (51.1%) reported a shifting perspective about learners and teachers' roles, while 40% reported intentions to critically explore and integrate GenAI tools into their learning and future teaching ([Appendix 4B](#)). These results show support for Design Principle 1 (critical reflections) and Design Principle 3 (Human–AI shared regulation). Furthermore, participants' responses provided additional empirical support for DP1, as illustrated by the following excerpts:

I have to update my knowledge according to the coming generation needs and technology literacy.

(Student 6)

... need to be flexible and positive in our mind and should always foresee our future... and prepare.

(Student 7)

Support for Design Principle 3 (Human–AI shared regulation) was also evident in students' reflections, which indicated an increasing capacity and intentionality to critically regulate their use of AI:

It is easier for me now to evaluate the balance of affordances and threats provided by these tools.

(Student 3)

Nevertheless, students' feedback indicates that Design Principle 2 requires further adjustment. In particular, the importance of practical, hands-on activities with GenAI tools was highlighted, reflecting that not only active learning but also opportunities for experiential learning are highly needed:

...workshop could be more extended in time, especially in the hands-on part, so that we would be able to see the tools in more detail or engage in collaborative tasks with the AI tools as well.

(Student 7)

Design Principle 2 was thus refined to include experiential learning through authentic tasks to develop AI understanding and application skills. While not yet common in higher education GenAI training, such approaches have reduced AI anxiety and increased readiness among younger learners (Förster et al., 2024; Su et al., 2023). We revised both the principle and its design embodiment to include extended hands-on time, self-evaluation of GenAI outputs to conclude on its strengths and limitations and an applied task involving lesson and material design (See Appendix 1E).

Principle 2. Active and Experiential Learning Approach—Training should offer hands-on opportunities for students to co-construct understanding and experiment with GenAI tools in authentic tasks.

Iteration 2: Evaluation of the AI literacy training programme and design principles

Changes in TAICS scores

Due to the ordinal and non-normal distribution of data, the Wilcoxon signed-rank test was used to assess changes in AI self-efficacy. As the analysis involved comparisons across multiple subscales, this study applied the Benjamini–Hochberg (BH) correction to address the risk of false-positives. Results showed a statistically significant increase in overall TAICS scores ($Z = -3.16$, $p < 0.001$), with median score rising from 3.79 (pre-test) to 4.42 (post-test) in the scale of 1–5. The effect size was large ($r = 0.61$), indicating a meaningful impact of the intervention.

Significant gains were also observed in subscales with large or medium-to-large effect sizes: AI knowledge ($p_{AIK} = 0.032$, $r_{AIK} = 0.41$), ethics ($p_{AIE} = 0.041$, $r_{AIE} = 0.39$), AI human-centered mindset ($p_{AIHu} = 0.032$, $r_{AIHu} = 0.57$), and AI professional development ($p_{AIPD} = 0.005$, $r_{AIPD} = 0.55$). However, improvements in AI pedagogy ($p_{AIP} = 0.081$) and AI assessment ($p_{AIAs} = 0.390$) were not statistically significant. After applying the BH corrections, most of

the significance levels remained unchanged; only the result of AI ethics shifted from significant to marginally significant. These findings suggest the intervention effectively enhanced overall AI competency self-efficacy, though AI pedagogical and assessment competencies may require longer-term or more targeted development. The results will be triangulated with self-reflections and lesson plan evaluations to further explore implications for enhancing training prototype (See [Appendix 5](#)).

Participant's rating of workshop experience

Participants rated the workshop positively across multiple dimensions. Engagement levels were high ($M=4.62$ on a scale of 1–5, $SD=0.75$) and the perceived usefulness of materials and discussions was similarly rated ($M=4.42$, $SD=0.88$), indicating strong alignment with learners' needs. The mental effort required was moderate ($M=3.65$), and the difficulty level of AI tools and content was rated as appropriate ($M=3.05$), suggesting the materials were accessible yet cognitively stimulating.

Self-reflections

Thematic content analysis of post-workshop self-reflections echoed findings from Iteration 1. Most participants (95.8%) expressed positive emotional reactions (e.g. 'happy', 'excited' and 'useful'), and key themes from workshop take away included the following: (1) Shifting perspectives; (2) New learning and teaching practices; (3) Intention to engage in continuous learning; (4) Need to develop Human-Centered Mindset; and (5) Communicate with different GenAI tools. These themes align with the AI competence framework and reinforce results from TAICS scores and lesson task performance. Illustrative quotes are presented in [Appendix 4](#).

Quality of lesson plans based on TPACK rubric

TE students used an average of 2.67 GenAI tools ($SD=1.17$) in lesson planning, with 54.17% using more than two. Compared to initial familiarity with only ChatGPT, hands-on training enhanced exploration, evaluation and integration of multiple GenAI tools for creating multimedia teaching materials and authentic classroom applications.

Two researchers and one teacher educator assessed students' GenAI-supported lesson plans using an adapted TPACK-based rubric ($\kappa=0.73$). Although TE students self-reported moderate to high AI competency in pedagogy and assessment, this was not consistently evident in their artefacts. Only PK1 ($M=3.00$, $SD=1.07$) and PK2 ($M=3.32$, $SD=0.78$) scored above 3 on a 4-point scale. PK3, PK4 and TPK5 ranged from 2 to 3, while other criteria scored below average, revealing limited application of modern pedagogies such as differentiation and inquiry-based learning. TPK6 and TPK7 received the lowest mean score ($M=1.64$), indicating basic tool usage (e.g. slides, quizzes) and lack of creative application of GenAI in teaching. Heavy reliance on AI-generated content also led to some low contextualization and originality (XR: $M=2.14$) (See [Appendix 6](#)).

Post workshop reflections of two teacher educators and TE students revealed challenges in critically evaluating AI-generated content, largely due to limited pedagogical expertise in innovative practices and time constraints: 31% TE students would like to learn more about AI pedagogy, assessment and tools for teaching ([Appendix 4B](#)). This also helps explain the results of the insignificant change in AI pedagogical and assessment competencies scores. While prompt and output checklists supported the creation of basic-standard lesson plans,

deeper pedagogical understanding is essential to guide effective prompting and assessment of GenAI outputs (Gurl et al., 2024). These findings highlight the importance of embedding self-regulated learning, metacognitive skills, modern pedagogy and authentic assessment into TE and AI competency training (Ng, Tan et al., 2024). Such integration enables TE students to use GenAI tools to enhance, rather than replicate, ineffective teaching practices. Consequently, Design Principle 3 was refined to emphasize scaffolding of self-regulated learning, metacognitive development and effective prompting to enhance the competency of evaluation of AI outputs. The gap between perceived and demonstrated AI pedagogy and assessment competence informed Design Principle 4: embedding authentic assessment to support critical evaluation of learning and GenAI outputs (See Table 1).

FINAL DESIGN PRINCIPLES OF AI COMPETENCY TRAINING FOR TE STUDENTS

Findings from two iterations support the practicality and effectiveness of the initial design principles in enhancing AI competency self-efficacy, positive and critical attitudes, stronger intentions to engage responsibly with GenAI and broader tool exploration in authentic tasks. Feedbacks from learners and insights from discrepancy emerged between self-reported competence and actual performance in lesson plans created with GenAI revealing a need for stronger scaffolding. As a result, we refined Design Principles 2 and 3 and introduced a new principle (Table 1). The conjecture map was updated accordingly (Figure 4).

DISCUSSION

This DBR aimed to conceptualize design principles and evaluate the effectiveness of an AI competency training workshop designed to enhance teacher education (TE) students' AI competency self-efficacy, mindset transformation and application of GenAI in lesson planning and learning material creation. The study's design principles and embodied implementation contribute significantly to both the practical integration of AI literacy training in TE and the ongoing discourse around GenAI in education.

TABLE 1 Final design principles for AI competency training for TE students.

#	Refined design principles (DPs)	Design principle specification
DP1	Critical reflections	Training should provide 'disorienting dilemma' to trigger critical reflections on roles and agency of students and teachers, future technology orientation, impact and ethics of AI
DP2	Active and experiential learning	Training should support students' hands-on opportunity to co-construct understanding and experiment with GenAI tools in discipline-specific authentic tasks
DP3	Design for human–AI shared regulation processes	Training needs to focus on developing students' self-regulation and prompting skills to enable reflective, responsible and effective interaction with AI systems
DP4	Authentic and formative assessment	Assessments should integrate robust authentic and formative assessment to promote TE students' critical engagement with GenAI and AI competency continuous development

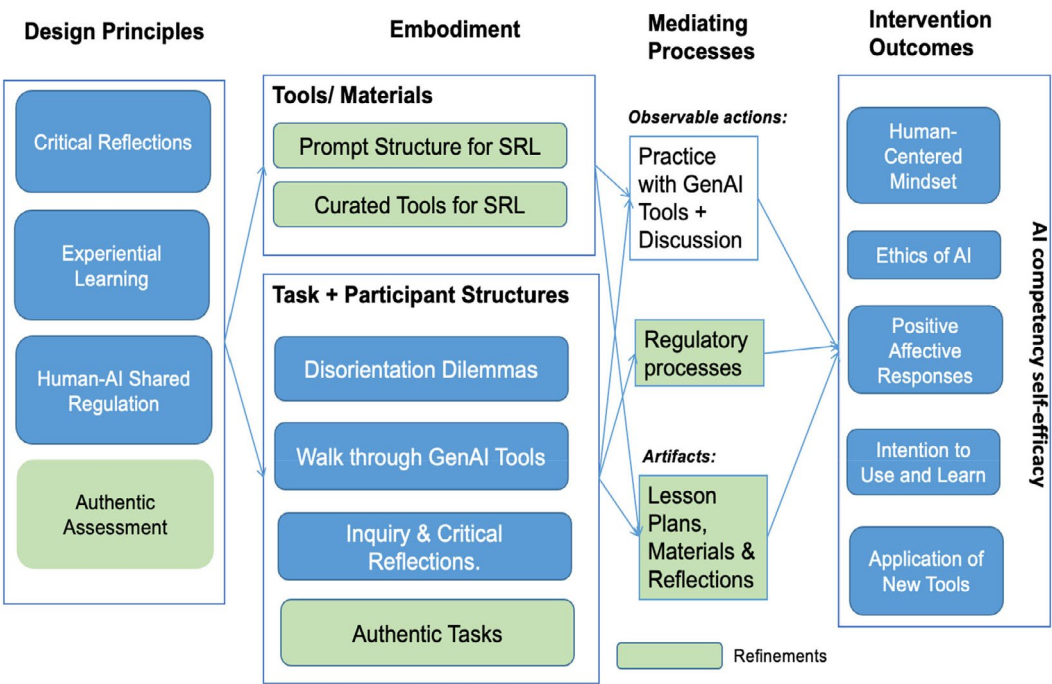


FIGURE 4 Final design principles and conjecturing map.

Research contributions to pedagogy and methods in AI literacy development

Findings confirm that the training design significantly enhanced TE students' AI self-efficacy, particularly in the areas of AI knowledge and ethics. These outcomes align with previous research from technically focused and longer AI literacy programmes in K–12 contexts (Markus et al., 2024; Ng, Wu et al., 2024). Remarkably, our training's emphasis on socio-emotional, cultural and ethical dimensions of GenAI significantly improved TE students' confidence and sense of professional competencies—domains often overlooked in current higher education AI literacy programmes (Chee et al., 2024). Reflections revealed increased engagement, a shift towards more critical and growth-oriented mindsets and deeper awareness of their roles as future educators in AI-enhanced environments. Such transformative experiences have been previously reported primarily within longer digital competency initiatives (Berg & Bozalek, 2023; Wang et al., 2014), underscoring the effectiveness and practicality of our proposed approach in the time constraint of TE students' education programmes.

Despite positive outcomes, gaps emerged between TE students' self-perceived AI pedagogical competence and their actual performance in TPACK-aligned lesson planning, likely due to limited prior pedagogical knowledge and time constraints of short workshops. Similar issues have also been reported in previous research, where GenAI use remained superficial due to insufficient prior knowledge, self-regulated learning skills and limited time (Abbas et al., 2024; Nguyen, Ilesanmi et al., 2024). To address this, our refined design principles emphasize formative, authentic assessment to build TE students' self-evaluation capacity and scaffold self-regulatory skills to support meaningful human–AI interaction, reduce over-reliance and promote metacognitive development (Fan et al., 2025; Molenaar, 2022a; Nguyen, Hong et al., 2024).

We propose an instantiated prototype of the refined design principles as a comprehensive AI competency training programme for TE students. This programme integrates self-regulatory

checklists (Cleary & Zimmerman, 2004), GenAI output evaluation rubrics and customized GPT tools designed to support self-regulated learning (Fan et al., 2025; Ng et al., 2024). Unlike generic tools, these pedagogically informed versions foster structured reflection and self-regulation. Supported by research in intelligent tutoring and adaptive learning systems (Azevedo et al., 2022; Yang et al., 2024), they promote feedback responsiveness and metacognitive development, preparing TE students to become lifelong, agentic learners.

This study addresses a critical gap in AI literacy training for TE students by developing and evaluating a targeted intervention that enhances both their theoretical understanding and practical skills in applying AI in their learning and pedagogical practice. First, it provides the first insight into empirical evidence supporting a comprehensive approach that integrates technical, affective, ethical and sociocultural dimensions, demonstrating significant gains in TE students' AI-related self-efficacy and mindset transformation. The transferable design principles from this research hold promise for broader adaptation in AI competency training programmes targeting various non-technical student groups.

Second, the study deepens AI literacy pedagogy by moving beyond active learning to include reflective, ethical and socio-culturally responsive practices. By embedding opportunities for TE students to question their assumptions, examine ethical dilemmas and reimagine their future roles, the training fosters both cognitive and affective dimensions of AI integration in higher education (Chee et al., 2024). This contribution advances ongoing debates on hybrid intelligence by positioning AI not as a substitute for professional judgement but as a socio-technical resource whose educational value depends on teachers' interpretive, ethical and pedagogical decision-making (Järvelä et al., 2024). In this sense, the study conceptualizes AI literacy as a form of agentic capacity building, where teachers learn to negotiate control, responsibility and meaning-making within human–AI configurations. By foregrounding reflection and ethical deliberation, the findings respond directly to concerns that current AI training risks instrumentalizing teachers' work and narrowing professional agency. Instead, the study demonstrates how carefully designed AI literacy interventions can strengthen teachers' agency with AI, thus enabling them to critically appropriate AI tools, align their use with pedagogical values and adapt them across diverse socio-cultural contexts.

Third, our research contributes to the advancement of evolving practices in AI competency assessment. While previous efforts have focused on self-report instruments (Carolus et al., 2023; Celik, 2023; Lee & Park, 2024), we align with Lintner (2024) in advocating for the integration of both qualitative and quantitative approaches. Our findings emphasize the importance of formative assessment, not only as a measure of learning but also as a mechanism for activating self-regulation and metacognitive reflection. This position is supported by recent case studies in AI literacy across disciplines. For example, Shalpegin and Nguyen (2024) emphasized the importance of formative feedback in GenAI-supported supply chain education. Similarly, Weber et al. (2025) demonstrated the role of intelligent feedback systems in legal writing instruction, while Yu et al. (2025) highlighted hybrid AI tools that promote student reflection through peer review and evaluation in online learning.

Moreover, we propose that transformative learning be recognized as a key outcome within the HASRL model (Järvelä et al., 2023), particularly when learners engage with GenAI through critical reflection. Our findings show that TE students moved beyond task-level regulation to questioning pedagogical beliefs, school system–student–teacher–AI role boundaries and future professional responsibility. This aligns with the emerging hybrid intelligence perspectives, wherein human oversight is critical to avoid cognitive offloading or automation dominance (Cukurova, 2025; Molenaar, 2022b). Integrating transformative learning within HASRL contributes to current discourse on agency in education by showing how GenAI can act as a catalyst for identity redefinition, positioning preservice teachers as active designers, evaluators and ethical decision-makers rather than passive recipients of AI outputs and technological disruption (Zhai, 2024).

Implications for practice and policy

The findings provide actionable implications for educators, curriculum designers and policy-makers seeking to integrate GenAI literacy into teacher education. Each implication aligns with the four final design principles, offering a coherent framework for practice and policy implementation.

TE programmes should embed structured opportunities for critical reflection on teachers' roles, ethics and agency in AI-mediated education. Institutional curricula can integrate reflective modules or discussion-based seminars that engage TE students with 'disorienting dilemmas' around GenAI use, fostering ethical awareness and future-oriented teaching identities (King, 2002). Policies supporting reflective and ethical AI engagement—as emphasized by UNESCO's AI competency framework—are essential for cultivating human-centered AI literacy in teacher preparation (UNESCO, 2024b).

AI literacy initiatives should adopt active and experiential pedagogies that allow TE students to experiment with GenAI tools through discipline-specific, authentic tasks such as lesson design, assessment planning or material creation. Teacher educators can adapt project-based learning and microteaching approaches to strengthen TPACK in AI-integrated contexts, building on established TPACK pedagogy (Lachner et al., 2021; Zhang & Zhou, 2023). Policymakers can encourage and resource experiential GenAI training within accredited teacher education programmes, ensuring equitable access to evolving AI tools.

To reduce overreliance on AI and build adaptive expertise, AI literacy programmes must scaffold SRL and effective human–AI interactions. This includes the development of metacognitive checklists, guided prompting strategies and rubrics for evaluating AI outputs (Fan et al., 2025; Ng, Tan et al., 2024). Teacher educators should model these practices to promote reflection-in-action and the cultivation of hybrid intelligence (Shalpegin & Nguyen, 2024). Policy frameworks could support professional development for faculty to implement SRL scaffolds and evidence-based AI integration strategies as suggested in O'Dea and Ng (2024).

Formative and authentic assessment should be institutionalized as mechanisms for evaluating TE students' AI competencies and reflective growth. Assessment policies can adopt performance-based evaluations—such as AI-supported lesson plans and reflective journals—to assess not only technical proficiency but also ethical reasoning and pedagogical adaptability. Assessment policy at programme level should be structured as ongoing development rather than one-time certification (Cukurova, 2025; Nguyen, 2025).

Collectively, these implications call for a systemic reframing of AI literacy as a continuous, cross-disciplinary curriculum effort that supports iterative skill development, aligning with prior recommendations (Chiu, 2024; Järvelä et al., 2024). Integrating the four design principles into curriculum and policy should advance not only technical competence but also ethical agency, pedagogical innovation and sustained self-regulation. Clearer policy guidance is also needed for ethical and SRL-aligned AI tool selection, and for ensuring access to evidence-based and privacy-protective platforms (Fan et al., 2025). Institutional policy should further clarify how AI literacy will be assessed, supported and scaled across programmes.

Limitations and future research

The study's small sample size and short intervention period limit the generalizability of results. Future research should replicate the training with larger and more diverse cohorts and explore its long-term effects. Longitudinal studies could provide insights into how sustained AI literacy development influences metacognitive, reflective and human–AI

interaction skills. Although the design prototype includes general AI tools and content that could generally support different subjects, how to develop the AI pedagogy and assessment methods for each specific subject could be further explored. Moreover, although small participant groups are typical in DBR due to the iterative and developmental nature of the approach, the voluntary participation in the first iteration may have introduced selection bias, as participants with higher motivation or prior interest in AI could be overrepresented.

Another limitation is that the design principles for AI competency training in teacher education could be conceptualized in alternative ways based on the theoretical grounding. Nevertheless, this study represents an initial attempt to integrate and synthesize relevant theoretical frameworks and translate them into design principles as emerging design theories, with the aim of facilitating their transfer into practical application.

Further research is also needed to design and evaluate customized GenAI and intelligent tutoring systems for TE. Promising examples from computer science (Yang et al., 2024) and legal education (Weber et al., 2025) suggest that these tools can support scaffolding for prompting, reflection and feedback generation. In addition, multimodal data collection and learning analytics can help identify optimal learner–AI interaction patterns. Studying help-seeking behaviours, reflection quality and self-monitoring during GenAI-supported tasks will deepen our understanding of effective and ethical human–AI interaction in education.

CONCLUSION

This DBR developed a set of design principles and evaluated the effectiveness of an AI literacy training programme aimed at enhancing TE students' AI competency self-efficacy, fostering a critical and growth-oriented mindset, encouraging continuous learning and supporting the practical application of GenAI in authentic learning tasks. The findings underscore the importance of strengthening TE students' self-efficacy, critical reflection and responsible use of AI in educational contexts through AI literacy training. It is essential for educators and policymakers to view AI competency as a dynamic, evolving process that demands ongoing support and iterative development. By building upon the human–AI Shared Regulation for Learning (HASRL) framework and refining AI pedagogy and assessment strategies, this study provides a foundation for future AI literacy initiatives—empowering TE students to become self-regulated learners today and ethical, innovative designers of AI-enhanced classrooms in the future.

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CONFLICT OF INTEREST STATEMENT

The authors have no conflict of interest to disclose.

DATA AVAILABILITY STATEMENT

The data are not publicly available due to privacy consent restrictions but can be accessed upon request and completion of data transfer agreements.

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ETHICS STATEMENT

The study adheres to the ethical guidelines from the Ethics Committee of Human Sciences at the University of Oulu. Informed consent was obtained from all participants prior to their involvement in the study.

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APPENDIX 1

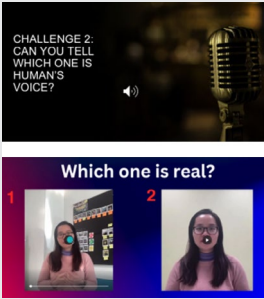
A. List of GenAI tools introduced in the workshop

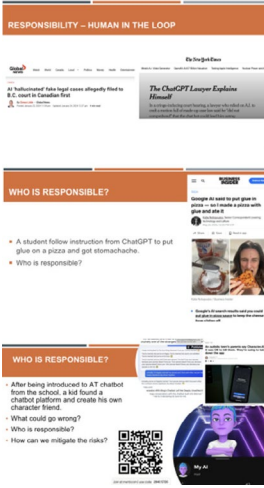
- AI assistant: ChatGPT, Microsoft Co-pilot, Magic School,
- Designing: [Canva.com](https://www.canva.com) (AI designer feature), Dall-E,
- Song creation: [Suno.com](https://www.suno.com),
- Infographic: [napkin.ai](https://www.napkin.ai)
- Video generation: [invideo.ai](https://www.invideo.ai)
- AI features available in current technology tools: Kahoot.co, [genially.com](https://www.genially.com)

B. Reflection questions

- How do you feel after the training?
- What are the key takeaways from this training?
- What do you want to learn more?

C. Detail tasks structures mapping with design principles: Task structures align with UNESCO AI competencies framework and initial design principles

Disorienting dilemma showcase	Example	Discussion prompts	Aligned with AI competence framework
GPT 4.o math tutoring capacity	https://www.youtube.com/watch?v=lvXZCocyU_M	What did you observe?	AI pedagogy and Assessment
GPT 4.o facilitation human interaction between father and son conversation	https://www.youtube.com/watch?v=dBrdd7xg-dg&ab_channel=KhanAcademy	How do the role of teachers change?	AI pedagogy
Deep fake identification quiz (voice, image, video)		What is the role of school? How assessments change? How could we prepare students and ourselves for this future?	AI ethics
Defeating AI detector demonstration	Live demonstration		AI ethics, AI pedagogy, AI assessment

AI biases cases			AI ethics, human-centered mindset
Human accountability cases		What could go wrong? How could we mitigate the risks?	Human-centered mindset

Disorienting dilemma showcase	Example	Discussion prompts	Aligned with AI competence framework
	Growth mindset game		AI professional development

D. Examples of embodiment of Design principle 2: Build human–AI interaction skill

COMPUTATIONAL THINKING FOR LESSON PLANNING:

- 
 - 1. Decompose:** What are the key parts of a lesson plan/problems? What are the key steps you usually do?
 - 2. Pattern seeking:** In each part what are the key content? Criteria for good lesson plans?
 - 3. Abstract:** Keep the most important aspects of a good lesson plan.
 - 4. Algorithm Design:** Decide on what tools to use, write. Instructions for the tools.

E. Example of embodiment of Design Principle 3: Active learning

YOUR CHALLENGE (EDUCATION)

- Design a Lesson Plan for a subject you teach that align with international and national standards, helping motivate learners, developing 21st century skills, values, and subject knowledge (Be creative to use AI or technology tools).
- Create creative/ engaging multimedia teaching materials for this lesson.
- Link to resources: <https://bit.ly/handoutfeb4>



Link to submit video, lesson plan and learning materials:
<https://bit.ly/feb4aiworkshop>

APPENDIX 2

INSTRUMENTS

TAICS scale—Effectiveness of training on the impact to the participants' AI competence was measured using the Teacher AI competence self-efficacy scale (TAICS) developed and validated by Chiu, Ahmad, and Çoban (2024) with $\alpha=0.971$. We chose TAICS as its current scale aligns with UNESCO's AI competency and TPACK frameworks. It consists of 24 items across six dimensions: AI Knowledge (AIK), AI Pedagogy (AIP), AI Assessment (AIAs), AI Ethics (AIE), Human-centered Education (AIHu) and AI Professional Engagement (AIPD), rated on a five-point Likert scale (1 = not at all; 5 = very much).

Learning experience—Learning experiences as an indication of practicality and effectiveness were evaluated using an adapted version of (Markus et al., 2024) training evaluation measure. Engagement, mental effort, difficulty level and overall learning experiences were rated on a five-point Likert scale (1 = not at all; 5 = very much). Effective training criteria required engagement scores above 3 and difficulty and mental effort scores between 2 and 3.

Self-reflection—Qualitative insights into emotional responses and shifts in participants' perspectives were gathered through reflective surveys based on Rolfe et al. (2001). Thematic (Bree & Gallagher, 2016) and content analyses (Kitchenham, 2006; Wood Daudelin, 1996) were employed to evaluate TE students' learning experiences.

Student's artefacts—To examine practical applications of AI competencies beyond self-report measures, students' lesson plans and materials were assessed. Evaluation employed a TPACK-based rubric adapted from Koh (2013) and Karakaya (2017) consisting of eight criteria: lesson completeness (PK-1), clear learning objectives (PK-2), suitability of teaching processes (PK-3), appropriateness of assessment (PK-4), suitable teaching materials (TPK-5), AI integration (AI-TPK-6), technology-content-pedagogy coherence (TPCK-7), and ethical-contextual awareness (XK-8). Ratings ranged from 1 (low mastery) to 4 (high mastery). Additionally, the number and variety of GenAI tools used by TE students were analysed as an indicator of competency translation.

APPENDIX 3

LESSON PLAN ASSIGNMENT RUBRIC

Content	Criteria	1	2	3	4
PK-1	Completeness of lesson plan (objectives, teaching materials, learning resources, teaching process)	Missing more than one part of the lesson plan	Missing one of the three components or all components are present but not suitable for the students. Does not include 21st-century skills objectives	Missing one component	All components are complete
PK-2	Completeness of objectives: knowledge skills, and attributes	Objectives are not aligned with content, duration, student group and lack clarity	Missing one of the three components or present but not suitable for students, content or unclear and not measurable/observable	Objectives are complete, clearly presented, measurable and aligned with students, content and assessment methods	Meets all criteria from level 3 and integrates 21st-century skills and AI competencies for students (e.g. technology application, creativity, critical thinking, evaluating AI-generated products, AI ethics, communication and collaboration)
PK-3	Completeness and suitability of the teaching process	Missing steps or lacking detailed descriptions of implementation, objectives and expected outcomes. The teaching process does not align with learning objectives or active learning methods	Missing a few parts or some activities do not align with objectives/learning methods/time allocation	The teaching process includes all four steps with complete components (objectives, content, expected outcomes, implementation guide). The four steps include: 1. Introduction/problem identification, 2. Knowledge formation/implementation, 3. Practice, 4. Application. Aligns with objectives. Mostly applies active teaching methods	The teaching process includes all four steps with complete components (objectives, content, expected outcomes, implementation guide). The four steps include: 1. Introduction/problem identification, 2. Knowledge formation/implementation, 3. Practice, 4. Application. Aligns with objectives and applies proven effective modern learning methods
PK-4	Appropriateness of assessment methods in organizing student learning activities	Assessment of learning products students must complete in each learning activity is NOT described	Assessment of student learning products is described but does not clearly specify assessment methods during the learning process	Assessment of student learning process and products is clearly described, including the criteria for student work. Does not yet consider AI impact in teaching and learning	Assessment methods are clearly described, realistic, and encourage students to apply 21st-century skills (creativity, technology application, critical thinking, teamwork, communication). Adjustments are made to align with AI impact in teaching and learning
TPK-5	Appropriateness of teaching materials and learning resources for organizing student activities (Including technology and AI use - Technology Selection)	Teaching materials and resources are not suitable for content, teaching methods, or student group. Not listed or unclear	Teaching materials and resources are somewhat aligned with learning products, however, student interaction (reading/writing/listening/viewing/doing) with these resources is NOT clearly described	Teaching materials and resources are aligned with learning products students must complete; student interaction (reading/writing/listening/viewing/doing) with these resources is clearly described	Teaching materials and resources are aligned with learning products students must complete; student interaction (reading/writing/listening/viewing/doing) with these resources is clearly described and aligns with active teaching methods. Students practice 21st-century skills (technology application, creativity, critical thinking, communication, and collaboration)

Content	Criteria	1	2	3	4
TPK-6	AI application in the lesson plan	No use of technology in teaching and supporting student learning. Technology use is inappropriate and does not aid student learning	Teacher uses technology (videos, games, etc), but AI tools are not included in the lesson	Teacher uses technology and AI tools in teaching (e.g. interactive activities with chatbots)	Students actively use AI for learning activities (e.g. using generative AI tools for group projects, creativity, or research)
TPCK-7	AI application suitability (alignment of content, learning objectives, teaching methods and technology)	Content, learning objectives, technology and teaching methods do not align in the lesson plan	Some alignment between content, learning objectives, technology and teaching methods in the lesson plan	Content, learning objectives, technology, and teaching methods are well-aligned in the lesson plan	Content, learning objectives, technology, and teaching methods are STRONGLY aligned in the lesson plan
XK-8	Ethical considerations and contextual awareness in AI and technology use	Text and language are unclear or inappropriate, direct copying of AI-generated prompts WITHOUT modification	Text and language are somewhat clear and appropriate but lack citations for sources and references	Text is revised for clarity and consistency, content is accurate and sources are cited. However, the lesson plan does not incorporate guidance on safe, effective and responsible AI use for students	Text is free of copied AI-generated prompts, revised for clarity and consistency, content is accurate, free of bias and properly cited. + The lesson plan integrates guidance on AI-assisted learning and ethical considerations in AI use

Theme	Description	Example Iteration 1	Example Iteration 2
New learning and teaching practices	Change in teaching methods and practices	'I learn that teachers need to motivate learners to WANT to learn, so they become active users of AI and technology rather than letting it do all the work for them'. (F13)	'I have determined the goals of using AI in teaching. Learned about teaching software and active teaching and learning'. (V26) 'What attracted me today was the approach to integrating AI into lessons in education'. (V4)
Intention to engage in continuous learning	Participants expressed strong interest in further exploring generative AI tools and incorporating them into their learning and practice teaching.	'We need to be flexible and positive in our mind and should always foresee our future and think what we can do to prepare for the possible future'. (F7) 'I would like to research more about the frameworks, for example, Unesco's that refer to AI competencies ... I also would like to try and answer more deeply those thought-provoking questions that were posed in the workshop, for example, "how does the role of schooling change" or "why do we need to learn ____ at school"'. (F1) 'I have to update my knowledge according to the coming generation needs and technology literacy. I learn that without AI technology, it's difficult to survive in the future' (F6)	'I will learn more about AI, its uses, and applications'. (V17) 'Try to study more and explore more to guide students in their learning. Always change oneself to ensure no one can replace you'. (V18) 'How to use some other AI tools and apply them to life'. (V4) 'I have learned a lot of useful things, and I want to apply AI to creating PowerPoint presentations'. (V5)
Human-centered mindset	Participants highlighted the importance of maintaining a human-centered approach to education, ensuring that AI serves as a tool to enhance, rather than replace, the human aspects of teaching and learning.	'Careful use of AI and not blindly trust on just AI And motivate myself that what am i doing will benefit me later not disappoint that if it not work now so not later so must proceed with growth mindset' (F12) 'I found very useful tools to support my teaching as well as my research work. I will definitely utilize these tools in the future! The rapid development of AI in terms of its resemblance with actual humans was surprising. It is easier for me now to evaluate the balance of affordances and threats provided by these tools, as I am more familiar with the possibilities (good and bad) that they create'. (F3) 'I will ... try to figure out first how I can incorporate AI first into my learning in a way that fits my needs and personality, and also figure out how I can deal with my own concerns about AI by learning more about it, before integrating it into my work'. (F1)	'Know how to use GPT Chat properly, know how to filter knowledge, and not rely too much on GPT Chat; know how to use GPT Chat for the right purpose'. (V1) 'I have gained more knowledge about AI and learned how to use AI in the teaching process, and how not to overuse it'. (V24) 'Understanding AI limitations, using AI as a supporting tool without reliance, and how to use AI effectively'. (V11) 'Effective applications of AI, limitations of AI, and not relying on AI'. (V25) 'Do not abuse AI or replace humans with it'. (V27)

Theme	Description	Example Iteration 1	Example Iteration 2
Interact with different GenAI tools	Enhance in awareness of different GenAI tools, their applications for learning and teaching as well as prompting skill	'I have learned about teaching software and active teaching tools'. (F2, 8) 'I liked the different tools and the overview of what they can do as it's always good to have more tools to use in future'. (F4) 'I like to learn about tools to both streamline as well as do better laborous parts, e.g. making slides, to free time for the planning and searching for pedagogical and domain content. Second, theoretical tools/frameworks (e.g. computational thinking) to apply when working with any GenAI tools to prepare for lessons. (F7)	'Useful and beneficial applications that I did not know before'. 'Learned about tools that assist in making presentations and slides, and how tools can create music and videos creatively and easily'. (V25) 'Learned how to write a complete prompt, create teaching materials that I didn't know how to create before'. (V29) 'Acquired beneficial knowledge such as creating videos from keywords using AI, creating lesson outlines'... (V5)

APPENDIX 5

WILCOXON SIGNED-RANK TEST AND EFFECT SIZES FOR THE PRE- AND POST-TEST SCORES OF TAICS AND SUBSCALES TAICS

Measure	Pre-test			Post-test			Z	p	p _{BH}	r
	Mean	SD	Mdn	Mean	SD	Mdn				
TAICS	3.76	0.78	3.79	4.12	0.91	4.42	-3.16	<0.001	0.01	0.61
AIK	3.49	1.03	3.5	3.88	1.04	3.75	-2.15	0.032	0.05	0.41
AIP	3.89	1.50	4.0	4.29	0.94	5.0	-1.75	0.081	0.09	0.34
AIAs	3.95	0.94	4.0	4.03	0.94	4.0	-0.86	0.390	0.41	0.17
AIE	3.89	0.89	4.0	4.19	1.01	4.5	-2.05	0.041	0.06	0.39
AIHu	3.63	0.78	3.5	4.23	1.04	4.75	-2.94	0.003	0.01	0.57
AIPD	3.73	0.78	3.75	4.14	0.98	4.0	-2.83	0.005	0.01	0.55

APPENDIX 6

DESCRIPTIVE STATISTICS OF LESSON PLAN'S SUB CRITERIA SCORES

Criteria	Mean	SD	Min	Max
PK1	3.0	1.069	1	4
PK2	3.32	0.780	2	4
PK3	2.32	0.945	1	4
PK4	2.05	0.899	1	4
TPK-5	2.32	0.945	1	4
TPK-6	1.64	0.492	1	2
TPCK-7	1.64	0.581	1	3
XK-8	2.14	0.834	1	3